

SYNPUC19CV mutation QC report

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Overview

This report summarizes detected sequence differences between a reference plasmid and a mutated/sample sequence. It is intended for research-use automation and visualization. Final biological interpretation should be reviewed by the researcher.

Plasmid information

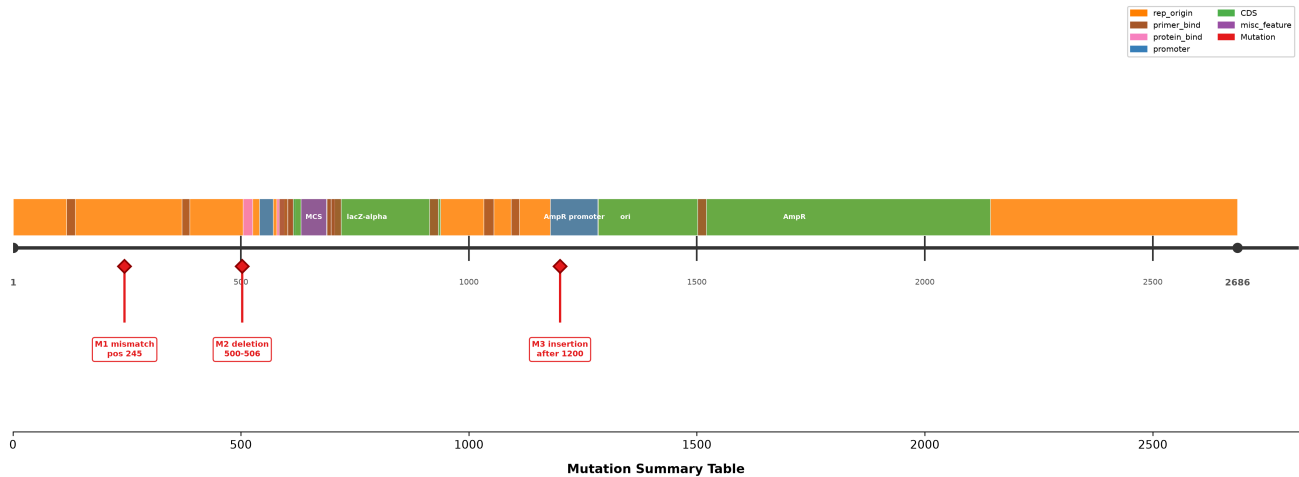
Field	Value
Plasmid name	SYNPUC19CV
Length	2,686 bp
Topology	circular
GenBank features	1
Reference GenBank	data/reference/reference_puc19.gb
Mutation CSV	outputs/tables/mutation_report.csv

Mutation summary

Type	Count
Total mutations	3
Mismatches	1
Insertions	1
Deletions	1
Other	0

Plasmid map

. — Mutation Map (3 mutations) (2,686 bp)



ID	Type	Position	Length (bp)	Note
1	mismatch	245	1	G changed to A at reference position 245
2	deletion	500-506	7	7 bp deleted from reference position 500-506
3	insertion	after 1200	6	6 bp inserted after 1200

Detected mutations

ID	Type	Position	Reference	Sample	Length
1	mismatch	245	G	A	1
2	deletion	500-506	TCACAAT	-	7
3	insertion	after 1200	-	TTGACA	6

Selected GenBank features

No selected GenBank features found.

Alignment preview

```
>reference_aligned
TCGCGCGTTTCGGTGATGACGGTGAAAACCTCTGACACATGCAGCTCCCGGAGACGGTCACAGCTTGTCTGTAAGCGGATGCCGGGAGCAGACAAGCCCGTCAGGGCGCG
... alignment preview truncated ...
```

Limitations

- Mutation positions are reported using the original/reference plasmid coordinate system.
- Insertions and deletions in repetitive regions may have ambiguous exact positions depending on alignment scoring.
- This script does not provide clinical or diagnostic interpretation.
- Large rearrangements may require manual review or a more advanced alignment workflow.